Nicolas Christianson

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nicochristianson.com

Academic Appointments

Johns Hopkins University. Baltimore, MD — starting August 2026
Assistant Professor, Department of Computer Science

Stanford University. Stanford, CA — 2025 – 2026
Postdoctoral Scholar, Management Science and Engineering
Mentors: *Ellen Vitercik and Ram Rajagopal*
Stanford Energy Postdoctoral Fellow

Education

California Institute of Technology. Pasadena, CA — 2025
Ph.D. in Computing and Mathematical Sciences.
Advisors: *Adam Wierman and Steven Low*
NSF Graduate Research Fellow
PIMCO Graduate Fellow in Data Science
Resnick Sustainability Institute (RSI) Scholar

Harvard College. Cambridge, MA — 2020
A.B. *summa cum laude* in Applied Mathematics.

Industry Collaborations

Microsoft Research — 2023 – 2025
Developed new algorithms to reliably deploy machine learning to power grid contingency analysis in collaboration with the Microsoft Research Special Projects group; resulted in a paper at L4DC 2025 (conference publication 8).

Amazon Prime Video — 2023 – 2024
Developed new algorithms for adaptive bitrate video streaming leveraging advancements in online optimization and learning. Yielded substantial improvements over the state-of-the-art and deployment to the Amazon Prime Video production environment, with results documented in a paper at SIGCOMM 2024 (conference publication 11).

Beyond Limits — 2022 – 2025
Developed algorithms for robust and efficient operation of real-world electricity/steam cogeneration resources in grids with high renewables penetration. Wrote a manuscript documenting results (journal paper 4) and incorporated the system model into SustainGym, an open-source library of sustainability-related benchmarks for reinforcement learning, documented in a paper at NeurIPS 2023 (conference publication 16).

Honors and Awards

ACM SIGEnergy Doctoral Dissertation Award	2026
Stanford Energy Postdoctoral Fellowship	2025
Ben P.C. Chou Doctoral Prize in Information Science and Technology	2025
Demetriades-Tsafka-Kokkalis Prize in Renewable Energy Sources	2025
PIMCO Graduate Fellowship in Data Science	2024
NSF Graduate Research Fellowship	2021
Phi Beta Kappa Junior 24	2019
John Harvard Scholarship	2017, 2019
Blair Research Fellowship (UPenn)	2018
Detur Book Prize	2017

Working Papers

** indicates equal contribution*
† indicates student I advised or coadvised

1. **Online Smoothed Demand Management**
   Journal version in preparation
   Adam Lechowicz, **Nicolas Christianson**, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
   Preliminary version accepted to ACM SIGMETRICS 2026 (conference publication 4)
2. **Online Conversion with Switching Costs: Robust and Learning-Augmented Algorithms**
   Journal version under review
   Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
   Preliminary version accepted to ACM SIGMETRICS/IFIP Performance 2024 (conference publication 14)
3. **Learning Dynamic Graphs, Too Slow**
   Preprint
   Andrei A. Klishin, **Nicolas H. Christianson**, Cynthia S.Q. Siew, Dani S. Bassett

Conference Publications

1. **Risk-Sensitive Peak-Aware Energy Scheduling: Competitive and Learning-Augmented Algorithms**
   ACM e-Energy 2026
   Finalist, Best Paper Award
   Lukas Himmelreich†, **Nicolas Christianson**, Adam Wierman
2. **Competitive and Learned Online Algorithms for HVAC Demand Management in Thermal Networks**
   ACM e-Energy 2026
   Adam Lechowicz, **Nicolas Christianson**, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
3. **Prediction-Specific Design of Learning-Augmented Algorithms**
   ACM SIGMETRICS 2026
   Sizhe Li†, **Nicolas Christianson**, Tongxin Li *Journal version:* POMACS 2026 (journal publication 1)
4. **Online Smoothed Demand Management**
   ACM SIGMETRICS 2026
   Adam Lechowicz, **Nicolas Christianson**, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
   Journal version: In preparation (working paper 1)
5. **Improving EV Aggregate Flexibility with End-to-End Learning**
   *Learning for Dynamics & Control Conference (**L4DC**) 2026, accepted*
   Apoorva Thanvantri†, Christopher Yeh, **Nicolas Christianson**, Adam Wierman
6. **Conformal Risk Training: End-to-End Optimization of Conformal Risk Control**
   *39th Annual Conference on Neural Information Processing Systems (**NeurIPS 2025**)*
   Christopher Yeh, **Nicolas Christianson**, Adam Wierman, Yisong Yue
7. **Learning for Online Scheduling with Competitive Fairness Guarantees**
   ACM e-Energy 2025
   Pengfei Li, **Nicolas Christianson**, Jianyi Yang, Adam Wierman, Shaolei Ren

8. **Fast and Reliable $N - k$ Contingency Screening with Input-Convex Neural Networks**
   *Learning for Dynamics & Control Conference (**L4DC**) 2025*
   Nicolas Christianson, Wenqi Cui, Steven Low, Weiwei Yang, Baosen Zhang
9. **Learning-Augmented Competitive Algorithms for Spatiotemporal Online Allocation with Deadline Constraints**
   ACM SIGMETRICS 2025
   Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
   Journal version: POMACS 2025 (journal publication 3)
10. **Risk-Sensitive Online Algorithms**
    *37th Annual Conference on Learning Theory (**COLT 2024**)*
    Nicolas Christianson, Bo Sun, Steven Low, Adam Wierman
11. **SODA: An adaptive bitrate controller for consistent high-quality video streaming**
    SIGCOMM 2024
    Tianyu Chen, Yiheng Lin, **Nicolas Christianson**, Zahaib Akhtar, Sharath Dharmaji, Mohammad Hajiesmaili, Adam Wierman, Ramesh K. Sitaraman
12. **Chasing Convex Functions with Long-term Constraints**
    *41st International Conference on Machine Learning (**ICML 2024**)*
    Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
13. **Online Algorithms with Uncertainty-Quantified Predictions**
    *41st International Conference on Machine Learning (**ICML 2024**)*
    Bo Sun, Jerry Huang[†], **Nicolas Christianson**, Mohammad Hajiesmaili, Adam Wierman, Raouf Boutaba
14. **Online Conversion with Switching Costs: Robust and Learning-Augmented Algorithms**
    ACM SIGMETRICS/IFIP PERFORMANCE 2024
    Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
    Journal version: Under review (working paper 2)
15. **The Online Pause and Resume Problem: Optimal Algorithms and An Application to Carbon-Aware Load Shifting**
    ACM SIGMETRICS/IFIP PERFORMANCE 2024
    Adam Lechowicz, **Nicolas Christianson**, Jinhang Zuo, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
    Journal version: POMACS 2023 (journal publication 5)
16. **SustainGym: Reinforcement Learning Environments for Sustainable Energy Systems**
    *37th Annual Conference on Neural Information Processing Systems (**NeurIPS 2023**), Datasets and Benchmarks Track*
    Christopher Yeh, Victor Li, Rajeev Datta, Julio Arroyo, **Nicolas Christianson**, Chi Zhang, Yize Chen, Mohammad Mehdi Hosseini, Azarang Golmohammadi, Yuanyuan Shi, Yisong Yue, Adam Wierman
17. **Pricing Uncertainty in Stochastic Multi-Stage Electricity Markets**
    *62nd IEEE Conference on Decision and Control (**CDC 2023**)*
    Lucien Werner*, **Nicolas Christianson***, Alessandro Zocca, Adam Wierman, Steven Low

18. **Optimal robustness-consistency tradeoffs for learning-augmented metrical task systems**
    *26th International Conference on Artificial Intelligence and Statistics (**AISTATS 2023**)*
    Nicolas Christianson, Junxuan Shen†, Adam Wierman
19. **Smoothed Online Optimization with Unreliable Predictions**
    ACM SIGMETRICS 2023
    Daan Rutten, **Nicolas Christianson**, Debankur Mukherjee, Adam Wierman
    Journal version: POMACS 2023 (journal publication 6)
20. **Dispatch-aware planning for feasible power system operation**
    *22nd Power Systems Computation Conference (**PSCC 2022**)*
    Nicolas Christianson*, Lucien Werner*, Adam Wierman, Steven Low
    Journal version: EPSR 2022 (journal publication 7)
21. **Chasing Convex Bodies and Functions with Black-Box Advice**
    *35th Annual Conference on Learning Theory (**COLT 2022**)*
    Nicolas Christianson, Tinashe Handina, Adam Wierman

Journal Publications

1. **Prediction-Specific Design of Learning-Augmented Algorithms**
   Proc. of the ACM on Measurement and Analysis of Computing Systems; vol. 10, iss. 1, art. 18, pp. 1-57, 2026
   Sizhe Li†, **Nicolas Christianson**, Tongxin Li
   Also appeared at ACM SIGMETRICS '26 (conference publication 3)
2. **End-to-End Conformal Calibration for Optimization Under Uncertainty**
   Transactions on Machine Learning Research, 2025
   Honorable Mention, Best Poster Award at LANL Grid Science Winter School 2025
   Christopher Yeh*, **Nicolas Christianson***, Alan Wu, Adam Wierman, Yisong Yue
   Preliminary versions appeared at Workshops on Tackling Climate Change with Machine Learning at ICLR '23 and NeurIPS '24 (workshop papers 2, 1)
3. **Learning-Augmented Competitive Algorithms for Spatiotemporal Online Allocation with Deadline Constraints**
   Proc. of the ACM on Measurement and Analysis of Computing Systems; vol. 9, iss. 1, art. 8, pp. 1-49, 2025
   Adam Lechowicz, **Nicolas Christianson**, Bo Sun, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
   Also appeared at ACM SIGMETRICS '25 (conference publication 9)
4. **Robust Machine-Learned Algorithms for Efficient Grid Operation**
   Environmental Data Science; vol. 4, art. e24
   Nicolas Christianson, Christopher Yeh, Tongxin Li, Mehdi Hosseini, Mahdi Torabi Rad, Azarang Golmohammadi, Adam Wierman
   Preliminary version appeared at NeurIPS '22 Workshop on Tackling Climate Change with Machine Learning (workshop paper 3)
5. **The Online Pause and Resume Problem: Optimal Algorithms and An Application to Carbon-Aware Load Shifting**
   Proc. of the ACM on Measurement and Analysis of Computing Systems; vol. 7, iss. 3, art. 45, pp. 1-32, 2023
   Adam Lechowicz, **Nicolas Christianson**, Jinhang Zuo, Noman Bashir, Mohammad Hajiesmaili, Adam Wierman, Prashant Shenoy
   Also appeared at ACM SIGMETRICS/IFIP PERFORMANCE '24 (conference publication 15)

6. **Smoothed Online Optimization with Unreliable Predictions**
   Proc. of the ACM on Measurement and Analysis of Computing Systems; vol. 7, iss. 1, art. 12, pp. 1-36, 2023
   Daan Rutten, **Nicolas Christianson**, Debankur Mukherjee, Adam Wierman
   Also appeared at ACM SIGMETRICS/IFIP PERFORMANCE '24 (conference publication 19)
7. **Dispatch-aware planning for feasible power system operation**
   Electric Power Systems Research; vol. 212: 108597, 2022
   Nicolas Christianson*, Lucien Werner*, Adam Wierman, Steven Low
   Also appeared at PSCC '22 (conference publication 20)
8. **Optimizing the human learnability of abstract network representations**
   Proceedings of the National Academy of Sciences; vol. 119, iss. 35: e2121338119, 2022
   William Qian, Christopher W. Lynn, Andrei A. Klishin, Jennifer Stiso, **Nicolas H. Christianson**, Dani S. Bassett
9. **Architecture and evolution of semantic networks in mathematics texts**
   Proceedings of the Royal Society A; vol. 476, iss. 2239: 20190741, 2020
   Nicolas H. Christianson, Ann Sizemore Blevins, Dani S. Bassett
10. **Structural and Functional Influence of the Glycine-Rich Loop G^{302}GGGY on the Catalytic Tyrosine of Histone Deacetylase 8**
    Biochemistry; vol. 55, iss. 48: 6718-6729, 2016
    Nicholas J. Porter, **Nicolas H. Christianson**, Christophe Decroos, David W. Christianson
11. **Biochemical and Structural Characterization of HDAC8 Mutants Associated with Cornelia de Lange Syndrome Spectrum Disorders**
    Biochemistry; vol. 54, iss. 42: 6501-6513, 2015
    Christophe Decroos, **Nicolas H. Christianson**, Laura E. Gullett, Christine M. Bowman, Karen E. Christianson, Matthew A. Deardorff, David W. Christianson

Workshop Papers

1. **End-to-End Conformal Calibration for Robust Grid-Scale Battery Storage Optimization**
   *Workshop on Tackling Climate Change with Machine Learning at **NeurIPS 2024***
   Christopher Yeh*, **Nicolas Christianson***, Adam Wierman, Yisong Yue
   Full version: TMLR (journal paper 2)
2. **Decision-Aware Uncertainty-Calibrated Deep Learning for Robust Energy System Operation**
   *Workshop on Tackling Climate Change with Machine Learning at **ICLR 2023***
   Christopher Yeh, **Nicolas Christianson**, Steven Low, Adam Wierman, Yisong Yue
   Full version: TMLR (journal paper 2)
3. **Robustifying Machine-Learned Algorithms for Efficient Grid Operation**
   *Workshop on Tackling Climate Change with Machine Learning at **NeurIPS 2022***
   Nicolas Christianson, Christopher Yeh, Tongxin Li, Mahdi Torabi Rad, Azarang Golmohammadi, Adam Wierman
   Full version: Environmental Data Science (journal paper 4)

Invited Talks and Seminars

End-to-End Learning for Uncertainty- and Risk-Aware Decision-Making
Stanford University - RAIN Seminar. February 2026

End-to-End Learning for Fast Contingency Screening
INFORMS Annual Meeting. October 2025

Reliable AI-Augmented Algorithms for Energy

ACM e-Energy '26 – Doctoral Dissertation Award Talk. June 2026
UC Berkeley – eMERGE Seminar. September 2025
Cornell University – ORIE Seminar. February 2025
MIT Sloan School of Management – OM Seminar. January 2025
Yale School of Management – Operations Seminar. January 2025
Johns Hopkins University – CS Seminar. December 2024
Columbia Business School – DRO Seminar. December 2024

Reliable ML-Augmented Algorithms for Energy and Sustainability

INFORMS Annual Meeting. October 2024
UMass Amherst – Systems and Sustainability Seminar. October 2024
Cornell ORIE – Young Researchers Workshop. October 2024
UC Berkeley – Energy Modeling, Analysis, & Control Group. September 2024

Learning-augmented algorithms for online optimization and beyond

Alberta Machine Intelligence Institute (Amii) AI Seminar. July 2024

Risk-Sensitive Online Algorithms

UMass Amherst – CS Theory Seminar. November 2025
INFORMS APS Conference. July 2025

Robust Machine-Learned Algorithms for Efficient Grid Operation

CAST Annual Program Review, Caltech. October 2023

Provable Guarantees on AI/ML for Metrical Task Systems and Classification

UMass Amherst – CS Theory Seminar. October 2023

Optimal Robustness-Consistency Tradeoffs for Learning-Augmented Metrical Task Systems

INFORMS Annual Meeting. October 2023

Chasing Convex Bodies and Functions with Black-Box Advice

Asilomar Conference on Systems and Signals. November 2022
Harvard University – Na Li's Research Group. October 2022
UMass Amherst – Data Science Deep Dive Seminar. October 2022
INFORMS Annual Meeting. October 2022

Research Mentorship

Sizhe Li (CUHK Shenzhen B.S. '26) 2025 – 2026
Topic: Learning-augmented online algorithms
Next step: Johns Hopkins CS PhD student (in my group)

Elizabeth Rogers (Harvey Mudd B.S. '26) 2025
Topic: Health-aware optimal power flow

Apoorva Thanvantri (Caltech B.S. '26) 2025
Topic: Improving electric vehicle aggregate flexibility with end-to-end learning
Next step: Harvard EE PhD student

Lukas Himmelreich (ETH Zurich M.Sc. '25) 2025
Topic: Risk-sensitive algorithms for peak-aware energy scheduling
Next step: ML engineer at Julius Baer

James Chen (Caltech B.S. '24) 2023 – 2024
Topic: Learning-augmented online optimization with ramp constraints
Next step: MIT EECS PhD student

Junxuan (Helen) Shen (Caltech B.S. '24) 2022 – 2024
Topic: Learning-augmented algorithms for multiserver convex function chasing
Next step: MIT EECS PhD student

Jerry Huang (Caltech B.S. '24) 2022 – 2023
Topic: Online algorithms with uncertainty-quantified predictions
Next step: CMU CS PhD student

Teaching Experience

California Institute of Technology

→ *Teaching Assistant*

CS 146: Control and Optimization of Networks — Spring 2024
CS 42: Computer Science Education in K-14 Settings — Winter 2024
CS 146: Control and Optimization of Networks — Winter 2023
CS 42: Computer Science Education in K-14 Settings — Winter 2023

Harvard College

→ *Peer Tutor, Harvard Bureau of Study Counsel/Academic Resource Center*

Math 25a: Theoretical Linear Algebra and Real Analysis I — Fall 2019
CS 181: Machine Learning — Spring 2019
APMTH 50: Introduction to Applied Mathematics — Spring 2019, Spring 2020
Math 132: Differential Topology — Spring 2019
APMTH 106: Applied Algebra — Fall 2018
STAT 110: Introduction to Probability — Fall 2018
CS 51: Intro to Computer Science II — Spring 2018, Spring 2019, Spring 2020
Math 25b: Theoretical Linear Algebra and Real Analysis II — Spring 2018

→ *Course Assistant*

Math Ma: Introduction to Functions and Calculus I — Fall 2017

Funding: Proposals and awards

Assisted in preparation of two proposals (PI: Adam Wierman) through Caltech's Center for Autonomous Systems and Technology, funded by Beyond Limits in 2022 and 2023. $230,000

NSF Graduate Research Fellowship, awarded 2021. $138,000

Professional Service

Workshop Organization

Learning-augmented Algorithms: Theory and Applications — 2024 – 2026
at ACM SIGMETRICS.

Physics-Informed Learning for Optimization and Control of Sustainable Energy Systems — 2026
at ACM e-Energy.

TPC Membership

ACM SIGMETRICS — 2027
ACM e-Energy — 2026
GreenSys Workshop — 2026

Journal Reviewing

Journal of the ACM — 2026
IEEE Transactions on Automatic Control — 2025, 2026
IEEE Transactions on Control of Network Systems — 2026

Operations Research 2025
IEEE/ACM Transactions on Networking 2023, 2024

Conference and Workshop Reviewing

→ *Conferences*

Asilomar Conference on Signals, Systems, and Computers. 2022, 2024
NeurIPS. 2025 (*"Top Reviewer"*), 2026
IEEE Conference on Decision and Control (CDC). 2025
Learning for Dynamics & Control Conference (L4DC). 2025, 2026
ACM SIGMETRICS. (subreviewer) 2022, 2023, 2024
ACM e-Energy. (subreviewer) 2023, 2024

→ *Workshops*

NeurIPS Workshop on Computational Sustainability. 2023

Grant Reviewing

Israel Science Foundation (ISF). 2026

Internal University Service at Caltech

Student Member of Caltech CMS AI/ML Admissions Committee 2022 – 2025
Caltech CMS Prelim Exam Preparation Coordinator 2021 – 2022

Outreach

Pasadena Public Schools. Science Night volunteer 2023 – 2025
Designing and organizing hands-on CS activities for elementary school students for "Science Nights" at public schools in and around Pasadena.

Caltech Accountability Partners Program. PhD application mentor 2022 – 2025
iSTEM Scholars. Summer research mentor 2021
Project SHORT. PhD application mentor 2020 – 2025

Work Experience

Microsoft Research. Redmond, WA Summer 2023
Research Intern
Intern in the Special Projects group, developed reliable machine learning methods to accelerate contingency analysis in energy grids while ensuring provable guarantees on performance.

KOACORE. Remote Spring 2023
Machine Learning Lead and Consultant
Led development of proof-of-concept and product strategy for KOA-SUPPLY, an AI-driven healthcare supply chain marketplace which has since been acquired by Stead Impact Ventures and now operates as Cato.

The Boston Consulting Group. Boston, MA Summer 2019
Summer Associate
Partnered with a top-10 global biopharmaceutical company to optimize its supply and manufacturing networks, using data and digital-driven techniques to forecast production needs and increase efficiency.

Covance. Princeton, NJ Summer 2017
Data Science Intern
Developed statistical and machine learning models to forecast clinical trial patient recruitment.